Bjørn Kåre Sæbø

# Topics in Control

Thesis for the degree of Philosohiae Doctor

Trondheim, August 2025

Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Engineering Cybernetics

NTNU
Innovation and Creativity

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ISBN 978-82-326-7040-6 (printed version)
ISBN 978-82-326-7039-0 (electronic version)
ISSN 1503-8181 (printed version)
ISSN 2703-8084 (electronic version)

Doctoral theses at NTNU, 2023:168
ITK report nr.: 2023-11-W

Printed by NTNU Grafisk senter

# Summary

This thesis presents results in task-priority control for underwater vehicle manipulator systems.

# Preface

This thesis is submitted in partial fulfilment of the requirements for the degree of Philosophiae Doctor (PhD) at the Norwegian University of Science and Technology (NTNU). The work has been carried out at the Department of Engineering Cybernetics (ITK), with Professor Kristin Y. Pettersen as the main supervisor and Professor Jan Tommy Gravdahl as co-supervisor. This work has been supported through the ERC advanced grant project CRÈME: Control of light vehicle-manipulator systems (grant no. 101017697).

## Acknowledgements

While writing a PhD is probably thought of by many as a lonely endeavor, I have not found it to be the case. During these years of writing the thesis, I have been fortunate enough to meet and work with many wonderful people, and I am hugely grateful to all of them. First and foremost, I must thank my supervisors Kristin and Jan Tommy.

During my visit to the Robotic Systems Lab at TU Wien, I was incredibly kindly received by professor Christian Ott and his group. A lot of my intuition for redundant robotic systmes is due to him.

Through my work at the applied underwater robotics laboratory I have been introduced to the practical side of robotics, which I think has greatly strengthened my understanding of the field as a whole. I would like to thank professors Asgeir Sørensen and Geir Johnsen for their infectious enthusiasm and drive to achieve real-life results, they have been a great motivation throughout these years. I would also like to thank dr. Pedro de la Torre, who has been the primus motor behind any subsea operations, be it in the Trondheimsfjord or at Svalbard. He has been a great influence and mentor in all the practical, logistical, and other challenges for

real-life operations.

I'm also very grateful to all the people I've met and worked with at NTNU, especially Jan Inge Dyrhaug, Erling Tveter, Mads Erlend Lysøe, Henrik Schmidt-Didlaukies, Erlend Basso, Marinna Wrzos-Kaminska... and most importantly, my office mate for these past five years Markus Iversflaten. This PhD would not have been the same without him.

Of course my parents deserve much of the credit for getting me to where I am today, pushing med to be curious and analytical from a very early age. Finally, I want to thank Maria, my fantastic girlfriend. I don't believe I could have completed this degree without her support. Thank you for pushing me to achieve what you knew I could do, forcing me to realize when I was unhappy and needed to improve. You have lifted me when I've been down, kept me grounded when I was too optimistic and been an endeless source of motivation, happiness, and joy. I love you.

*13th November 2025, Year, Trondheim*
*Bjørn Kåre Sæbø*

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# Introduction